

# Multistream LSTM for Artifact Detection in Impedance Cardiography

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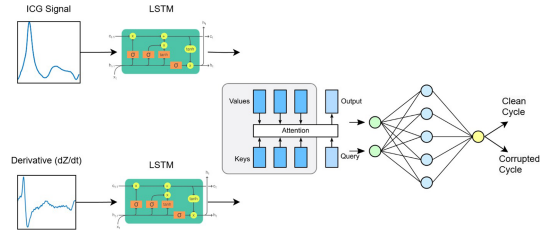
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**Abstract**—Monitoring cardiac hemodynamic parameters, such as cardiac output and pre-ejection period, is critical for assessing cardiovascular function, particularly in critically ill patients. Impedance cardiography (ICG) offers a noninvasive approach to measuring these parameters; however, its utility is often compromised by motion artifacts and electrode displacement. Many traditional artifact detection methods rely on rigid waveform templates, which may struggle to adapt to individual variations in ICG morphology, potentially resulting in limited generalization and higher misclassification rates in certain scenarios. In this study, we propose a deep learning-based framework that combines a multistream long short-term memory (LSTM) network, attention mechanisms, and ensemble learning to automatically detect corrupted ICG cycles. The model concurrently processes raw ICG signals and their derivatives to capture both temporal dynamics and morphological transitions. Attention layers highlight diagnostically relevant regions, while data augmentation and ensemble postprocessing improve generalization and robustness. The proposed method was validated on a dataset of 2000 ICG cycles from 20 individuals, achieving an accuracy of 96.42% against human expert visual detection, significantly outperforming traditional methods and single-stream LSTM models. This method enhances artifact detection and supports more reliable noninvasive cardiac monitoring.



**Index Terms**—Sensor applications, artifact detection, deep learning, impedance cardiography (ICG), long short-term memory (LSTM).

## I. INTRODUCTION

Cardiac hemodynamic parameters, such as stroke volume, cardiac output, and pre-ejection period, are critical for diagnosing and managing conditions, such as heart failure and coronary artery disease [1]. Traditional methods, including thermodilution pulmonary artery catheters, provide accurate measurements but are invasive and risky. Noninvasive techniques, such as Doppler echocardiography and MRI, offer safer alternatives but are costly and unsuitable for real-time monitoring. Impedance cardiography (ICG) has emerged as a noninvasive method to monitor cardiac parameters, addressing many limitations of traditional techniques [2], [3]. By measuring voltage changes and impedance, ICG provides insights into cardiovascular function, but it is sensitive to motion artifacts and electrode displacement, and requires expertise to manage corrupted cycles [4].

Efforts to improve ICG performance, such as advanced electrode designs [5], ensemble averaging [6], digital filtering [7], and wavelet-based noise reduction [8], have shown some success but often fail to handle motion artifacts effectively. Adaptive least squares methods using reference signals [9], [10] require additional sensors, limiting clinical applicability. Template-based approaches [11], [12], [13], for corrupted cycle detection show promise but struggle with individual

variability. Therefore, further development of automatic detection algorithms is necessary to accurately and efficiently identify corrupted ICG cycles, maximizing the potential of ICG as a noninvasive method for routine monitoring of hemodynamic parameters.

In this letter, we propose a novel approach to artifact detection in ICG signals using a multistream long short-term memory (LSTM) network. The method includes several preprocessing steps, such as filtering, segmentation, and data augmentation, to enhance signal quality and improve model generalization on limited data. The network processes both raw ICG signals and their derivatives through parallel input streams, leveraging LSTM layers, and attention mechanisms to capture temporal dependencies and sharp transitions effectively. An ensemble postprocessing approach further enhances robustness. Our method is preliminarily evaluated on ICG data from 20 individuals, demonstrating its potential to improve artifact detection and enable reliable noninvasive cardiac monitoring.

## II. MATERIALS AND METHODS

### A. Dataset

At the SRI Human Sleep Research Laboratory, polysomnography recordings were conducted on 20 healthy adults (ten female, average age  $46.2 \pm 8.4$  years) in sound-attenuated, temperature-controlled rooms. ECG signals were collected at 512 Hz using Ag/AgCl Meditrace electrodes, while ICG  $dZ/dt$  signals were recorded with a 4-lead,

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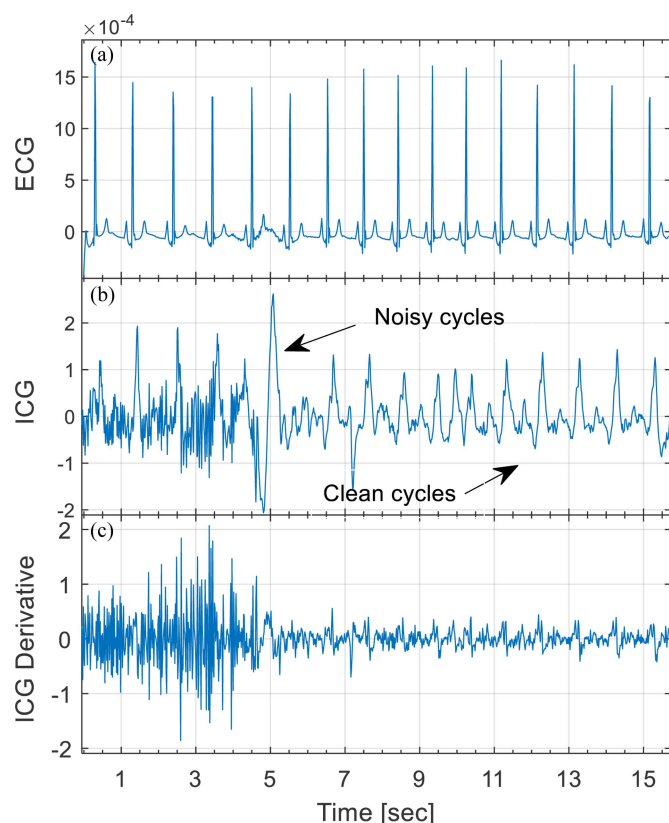


Fig. 1. Examples of simultaneously recorded ECG, ICG, and ICG derivative signals. (a) ECG. (b) ICG signal with distinct noisy cycles on the left and clean cycles on the right. (c) ICG derivative ( $dZ/dt$ ), where the high-frequency noise corresponds to the noisy cycles in the ICG signal.

8-point configuration via an HIC-4000 bioelectric impedance system. The inner electrodes were placed at the neck and lateral thorax, with outer electrodes positioned 3 cm above and below. The study was approved by the SRI Institutional Review Board, and all participants provided written consent. Examples of simultaneously recorded ECG, ICG, and ICG derivative signals are illustrated in Fig. 1.

## B. Preprocessing

1) *Filtering*: Initially, to enhance the quality of the collected ICG signals, a fourth-order Butterworth bandpass filter was employed to retain only frequencies between 0.5 and 35 Hz. This filtering process effectively eliminated both low-frequency drift and high-frequency noise, thus improving the accuracy of subsequent analyses.

2) *Cycle segmentation*: Next, the R peak in the ECG signal was identified as a cut-off point for the ICG signal, thereby dividing the signal into individual ICG Cardiac cycles. Since cycles varied in length, bilinear interpolation was applied to standardize them to 256 samples each, ensuring uniformity for model training and comparison.

3) *Augmentation*: Synthetic data augmentation was applied to improve model robustness. Controlled Gaussian noise with a mean of 0 and a standard deviation of 0.01–0.03 was added to simulate mild artifacts. In addition, time-domain jittering was introduced by randomly shifting the signal by 1%–3% of its length, mimicking electrode placement variations. These augmentations increased the training set size by 10 times and enhanced the model's generalization capabilities.

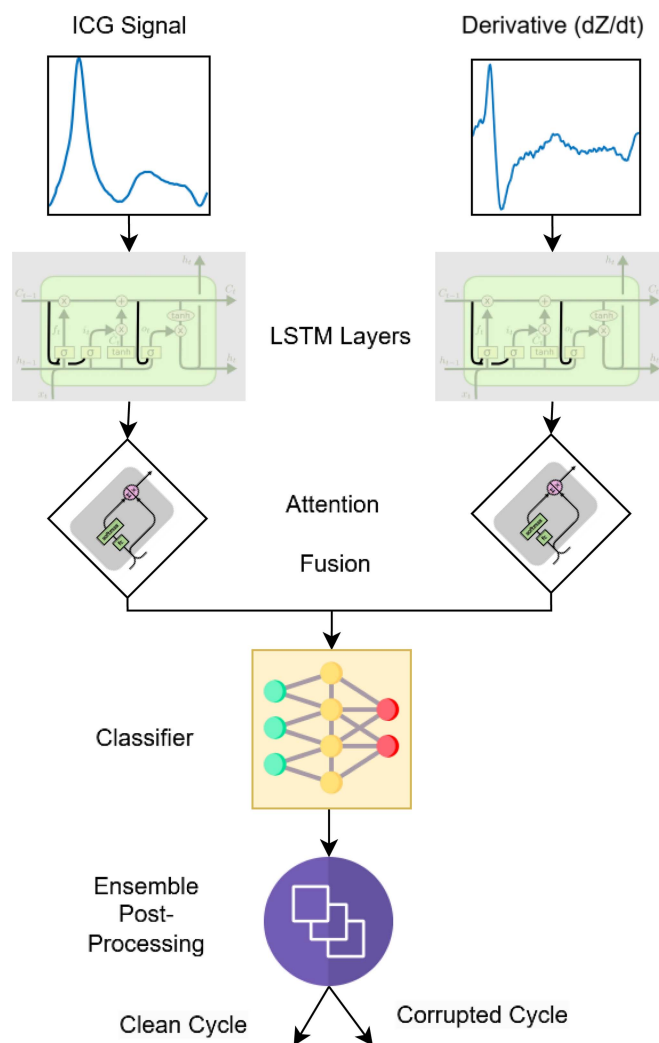


Fig. 2. Diagram of the proposed multistream LSTM model for artifact detection. The model processes ICG signals and derivatives through LSTM layers, applies attention mechanisms, and combines predictions via SoftMax and ensemble postprocessing to classify clean versus corrupted cycles.

4) *Normalization*: Finally, to further normalize the data, all ICG signal sampling values were divided by a normalizing parameter  $K$ , which was defined as the median of the amplitude of each ICG cardiac cycle. This step ensured that the data were scaled appropriately, as  $K$  accounted for variations in the amplitude of different cycles.

## C. Multistream LSTM Network Architecture

The proposed artifact detection algorithm is based on a multistream LSTM network, designed to process multiple aspects of the ICG signal concurrently, enhancing the model's capacity to capture complex temporal dependencies and morphological features (see Fig. 2).

1) *Input streams*: The network architecture incorporates two parallel input streams to improve the model's ability to capture critical features. The Raw ICG Stream processes the original ICG signal, allowing the network to extract high-level temporal dependencies across the entire cycle. Simultaneously, the derivative stream processes the first derivative of the ICG signal ( $dZ/dt$ ), highlighting rapid changes in the signal that correspond to key physiological events, such as the B and X points. This dual-stream approach enhances the model's sensitivity

to both long-term patterns and sharp transitions in the signal, leading to more accurate detection of artifacts.

2) *LSTM layers*: Each input stream feeds into a stack of two LSTM layers, each with 128 hidden units. The dual-layer configuration allows the network to capture both short-term and long-term dependencies within the signal. The LSTM layers employ peephole connections, allowing the cell state to directly influence gating mechanisms, providing the model with additional temporal context.

3) *Attention mechanism*: An attention mechanism is integrated after the LSTM layers, enabling the model to dynamically focus on critical segments of the ICG signal. The attention weights are computed based on the relevance of each time step to the artifact detection task, allowing the network to prioritize important features while downplaying irrelevant ones.

4) *Fusion layer*: The outputs from the parallel LSTM layers are concatenated and passed through a fusion layer, which integrates the information from the raw and derivative streams. This fusion process involves a fully connected layer with 256 neurons, followed by batch normalization to stabilize learning and reduce overfitting.

5) *Classification layer*: The fused feature vector is passed through a final classification layer, consisting of two fully connected layers (128 and 64 neurons, respectively) and a SoftMax output layer. Dropout regularization with a rate of 0.5 is applied between layers to mitigate overfitting.

6) *Ensemble postprocessing*: To further enhance the robustness and accuracy of artifact detection, an ensemble learning approach was adopted. The final prediction for each ICG cycle was obtained by averaging the outputs of three independently trained multistream LSTM models, each initialized with different random seeds. This ensemble method reduces prediction variance and enhances generalization across diverse test scenarios.

7) *Hyperparameter tuning and optimization*: Hyperparameters were optimized using a combination of grid search and Bayesian optimization, where the latter probabilistically selected the most promising hyperparameter sets based on prior evaluations. This approach efficiently explored the search space, tuning key hyperparameters such as the learning rate (0.0001–0.01), batch size (16, 32, and 64), and LSTM hidden units (64, 128, and 256). The final selected hyperparameters included a learning rate of 0.001, a batch size of 32, and 128 hidden units in the LSTM layers. The Adam optimizer with a cyclical learning rate policy was used, allowing the learning rate to oscillate between specified bounds, promoting exploration during training and helping to avoid local minima. The model was trained for up to 100 epochs, with early stopping based on validation loss to prevent overfitting.

### III. RESULTS

#### A. System Setting

The proposed method was implemented using TensorFlow, with training conducted on an NVIDIA Tesla V100 GPU. The input data for each cycle was structured as [batch size  $\times$  sequence length], where the sequence length was fixed at 256 for all experiments, while the batch size was treated as a hyperparameter.

#### B. Performance Analysis

A total of 2000 ICG cardiac cycles (100 per individual) were randomly selected and independently evaluated by an expert, who was blinded to the algorithm's classifications. Each cycle was labeled as either clean or corrupted based on visual inspection. Clean cycles were

TABLE 1. Performance Comparison of Different Model Configurations for ICG Artifact Detection

| Model Configuration                           | Accuracy | Sensitivity | Specificity | Precision | AUC   |
|-----------------------------------------------|----------|-------------|-------------|-----------|-------|
| Activity index-based [11]                     | 86.50%   | 73.35%      | 90.36%      | 69.09%    | NA    |
| Features + RF                                 | 88.15%   | 87.25%      | 89.05%      | 88.00%    | 0.881 |
| Feature + SVM                                 | 89.70%   | 88.90%      | 90.45%      | 89.30%    | 0.895 |
| Single-stream LSTM                            | 92.85%   | 90.45%      | 93.31%      | 91.45%    | 0.910 |
| Multi-stream (MS) LSTM                        | 93.35%   | 91.45%      | 94.12%      | 92.23%    | 0.925 |
| MS-LSTM + attention                           | 94.42%   | 92.65%      | 95.21%      | 93.33%    | 0.932 |
| MS-LSTM + attention + ensemble                | 95.25%   | 93.11%      | 95.92%      | 94.12%    | 0.945 |
| MS-LSTM + attention + ensemble + augmentation | 96.42%   | 95.10%      | 96.75%      | 95.87%    | 0.972 |

identified as those exhibiting the typical ICG waveform morphology, including clearly discernible and artifact-free B, Z, and X points. A total of 454 out of 2000 ICG cycles were scored by the expert as corrupted cycles.

The robustness of the proposed multistream LSTM model was evaluated using leave-one-subject-out cross-validation. In this approach, data from one subject were used as the test set, while the remaining data from 19 subjects were split into 90% for training and 10% for validation. This ensured that the model was tested on entirely unseen subjects, providing a rigorous assessment of its generalization capability. During each fold, early stopping was applied based on the validation loss calculated from the 10% validation set, optimizing the model's performance and preventing overfitting. Test set evaluations were conducted in a blinded manner, with predictions compared to expert annotations.

The proposed model achieved an overall accuracy of 96.42%, with a sensitivity of 95.10%, specificity of 96.75%, and an area under the curve (AUC) of 0.972, significantly outperforming the traditional Activity Index-based algorithm and single-stream LSTM (see Table 1). For comparison, we also evaluated conventional machine learning models, including support vector machine (SVM) and random forest (RF), which rely on manually extracted features including mean, standard deviation, entropy, energy, skewness, kurtosis, peak-to-peak amplitude, signal magnitude area, root mean square, and zero-crossing rate. These models achieved moderate accuracy (88.15% and 89.70%, respectively) while maintaining relatively low computational complexity. However, their dependence on handcrafted features limits their ability to capture complex temporal dependencies in noisy data. In contrast, the proposed LSTM-based approach learns directly from raw signals, enhancing performance without manual feature engineering.

Deep learning models demonstrated substantial performance gains but at the cost of higher complexity and memory demands. The single-stream LSTM, with an accuracy of 92.85% (AUC 0.910), introduced greater memory requirements due to its need to store weights and manage hidden states over sequential data. The multistream LSTM further improved accuracy to 93.35% (AUC 0.925) by processing multiple data streams in parallel, increasing computational overhead and memory usage compared to the single-stream variant.

By integrating attention mechanisms, we achieved an improved accuracy of 94.42% (AUC 0.932). This enhancement, while requiring additional memory during training to manage attention weights, unlocked an important performance leap. Further refinements, including ensemble postprocessing and data augmentation, propelled our model



to an outstanding peak accuracy of 96.42% (AUC 0.972). These powerful techniques, which involved training multiple models and expanding the dataset, primarily increased computational costs during the training phase. Importantly, at inference time, the model maintains its efficiency.

### C. Comparison and Discussion

In comparison, Forouzanfar et al. [12] utilized an activity index-based method for artifact detection, achieving 86.5% accuracy, 73.35% sensitivity, and 90.36% specificity. Our model shows an improvement of 9.92% in accuracy, 22.75% in sensitivity, and 6.39% in specificity, highlighting the advantage of the LSTM architecture in capturing complex temporal dependencies in the signal.

Sheikh et al. [14] employed an adaptive least-squares method focused on respiratory artifacts, achieving 93.18% sensitivity. While their approach targets specific artifact types, our model's sensitivity of 95.10% suggests it is better equipped to handle a broader range of artifacts.

In contrast, methods like Chabchoub et al. [8], which used discrete wavelet transforms to filter noise from ICG signals, and Pandey et al. [9], which employed adaptive filtering, focus on noise suppression rather than artifact detection. While effective at enhancing signal quality, these approaches do not detect or exclude entire corrupted cycles, a key advantage of our model.

Compared to traditional methods that rely on noise suppression or rigid templates, our approach offers a more flexible and robust solution for detecting corrupted cycles. While the derivative of the ICG waveform in Fig. 1 exhibits some fluctuations, the proposed method remains robust as it leverages both the ICG signal and its derivative, allowing the model to extract meaningful patterns despite local variations. In addition, filtering techniques, such as the Savitzky–Golay filter [15], could be explored to further refine the derivative signal and enhance performance.

Traditional machine learning models like SVMs and RFs depend on manual feature extraction, which often fails to capture complex temporal dependencies and nonlinear patterns in noisy ICG signals. These limitations lead to poor generalization and higher misclassification rates in the presence of individual variations in ICG morphology. In contrast, LSTM networks automatically learn these patterns directly from raw data, capturing both temporal dependencies and morphological variations. This reduces the need for feature engineering and enhances generalization across varying signal conditions, further demonstrating the strengths of the multistream LSTM model, particularly with attention mechanisms and data augmentation. Future work will involve validating these improvements on larger and more diverse datasets. In addition, methods such as CNN-based approaches, like those proposed by Mondal et al. [16] for efficient ECG signal quality assessment, could be explored to optimize the model for resource-constrained devices, enhancing its applicability in real-time clinical settings.

## IV. CONCLUSION

This study presented a novel multistream LSTM-based approach for detecting artifacts in ICG signals, demonstrating significant improvements in accuracy and generalization within the tested dataset compared to traditional methods. The algorithm's ability to detect and exclude corrupted cycles without requiring individualized templates highlights its potential as a valuable tool in noninvasive cardiac monitoring.

The integration of attention mechanisms and ensemble learning further enhanced the model's robustness, making it a promising foundation for future clinical applications. Future work will focus on expanding the dataset to include more subjects, particularly those with cardiovascular conditions. Additionally, networks such as CNNs will be incorporated alongside LSTMs for feature extraction. Efforts will also be directed toward adapting the model for deployment in real-time monitoring settings, addressing the complexities and challenges of such environments.

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